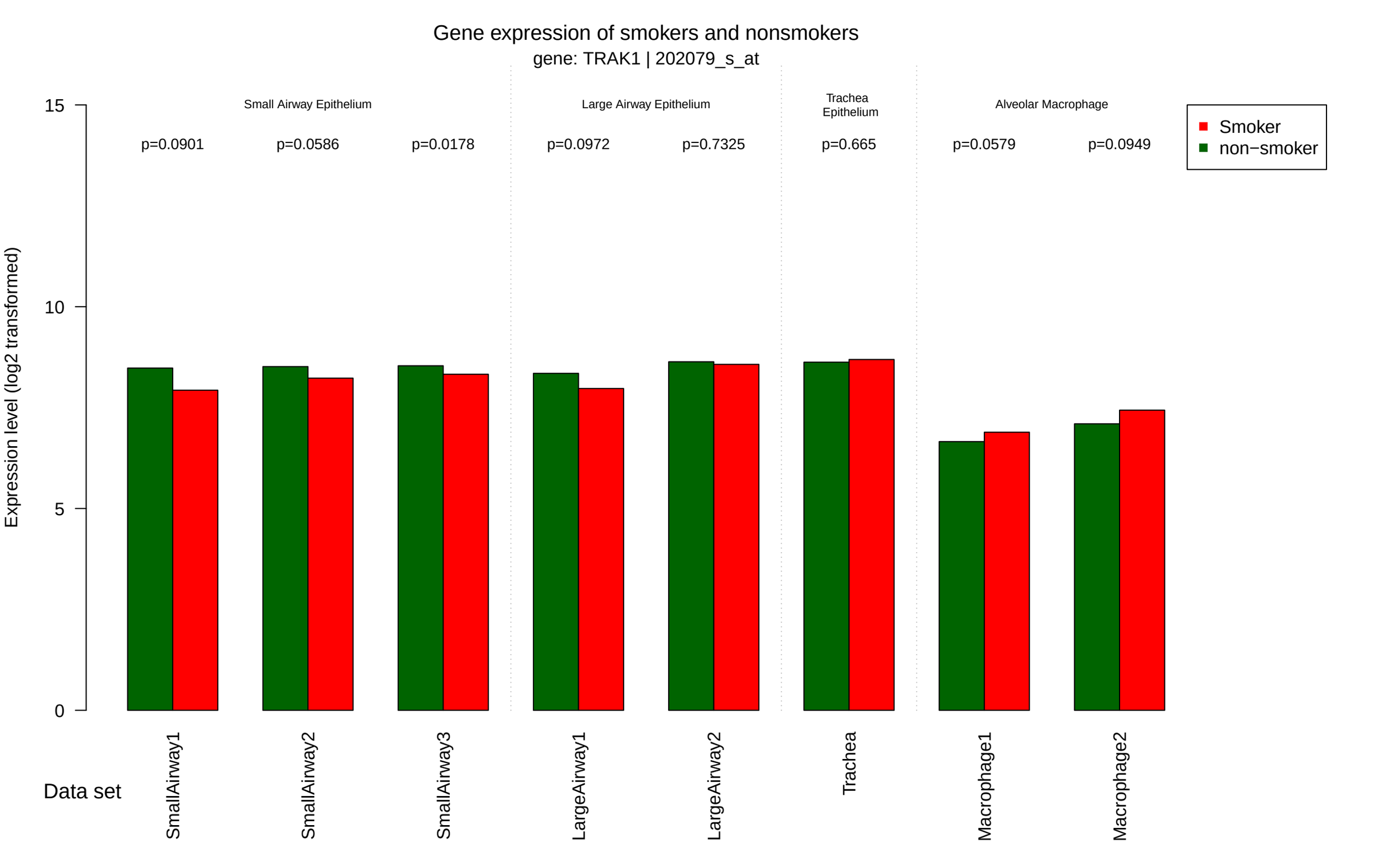
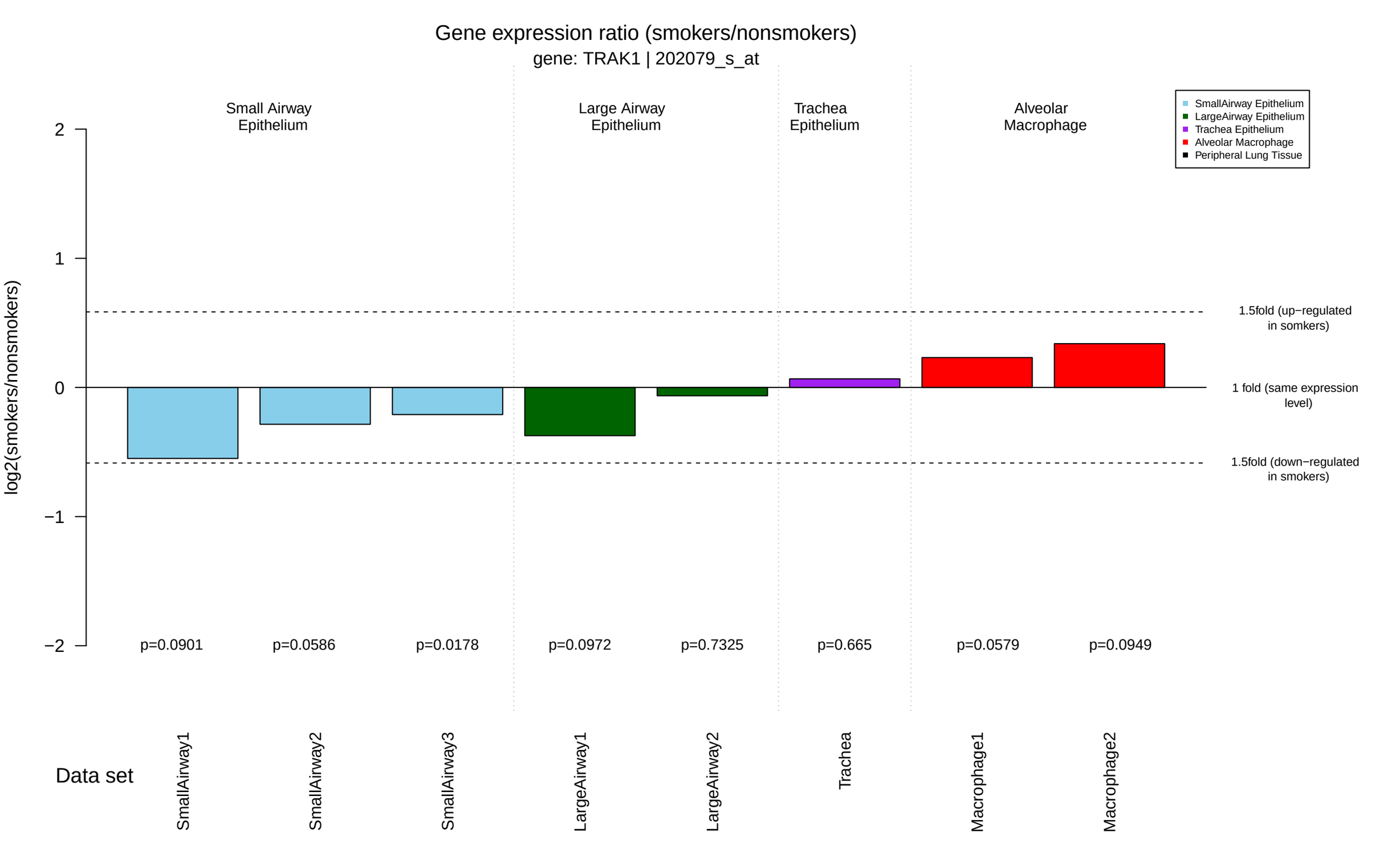
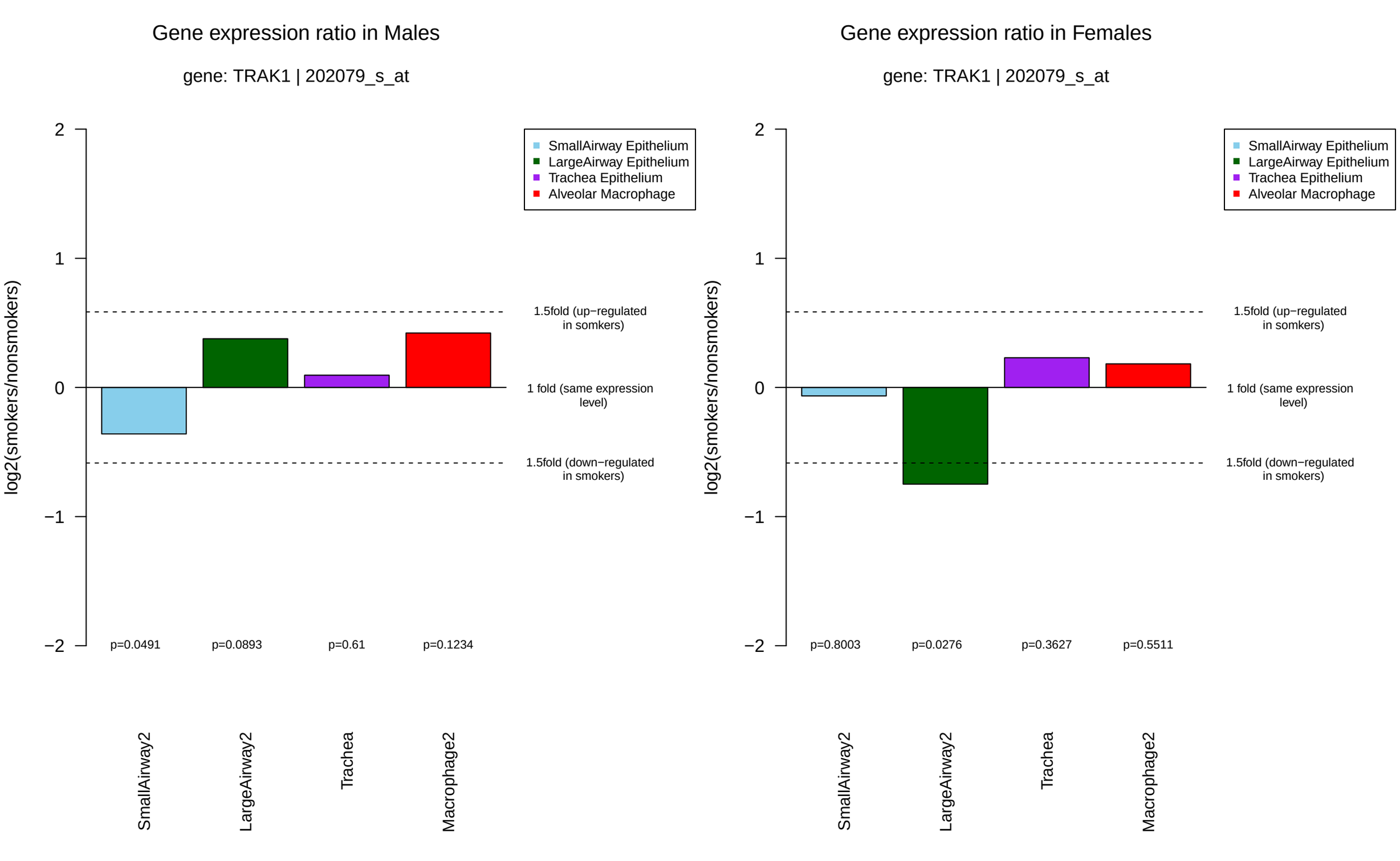
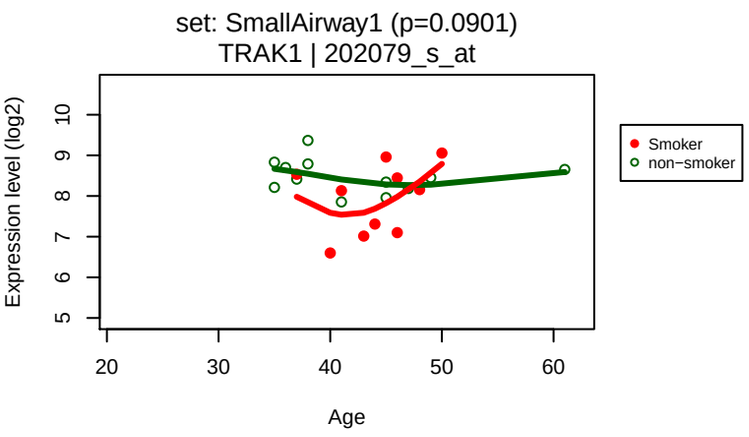
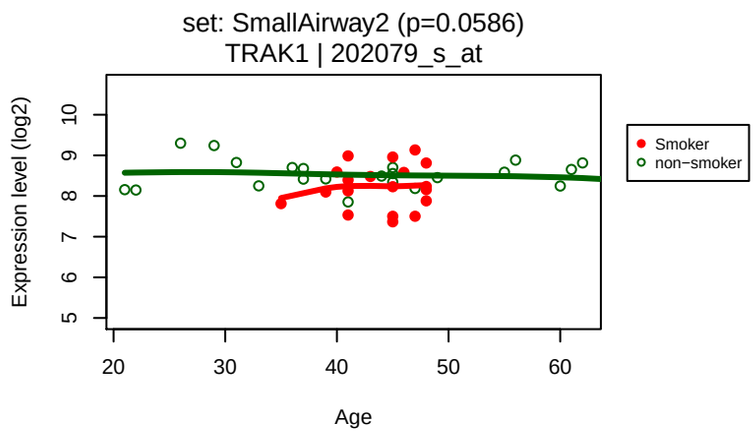
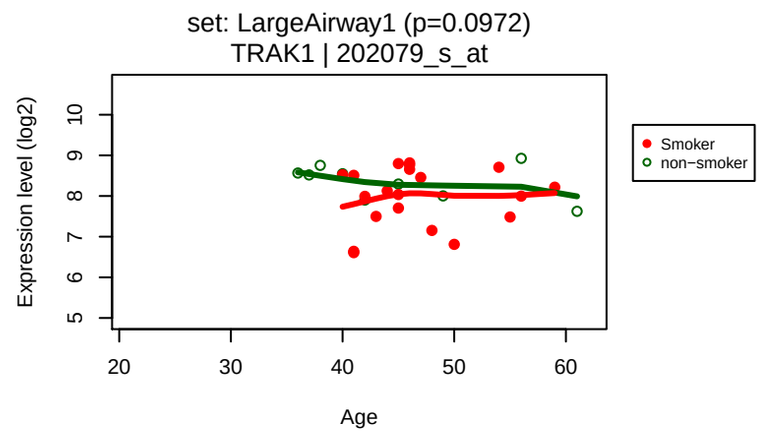

Gene expression of smokers and nonsmokers
gene: TRAK1 | 202079_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0901
p=0.0586
p=0.0178
p=0.0972
p=0.7325
p=0.665
p=0.0579
p=0.0949
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

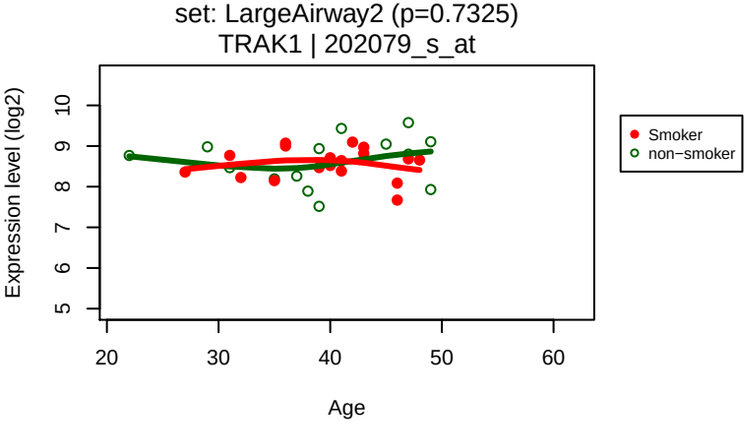

Gene expression ratio (smokers/nonsmokers)
gene: TRAK1 | 202079_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0901
p=0.0586
p=0.0178
p=0.0972
p=0.7325
p=0.665
p=0.0579
p=0.0949
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

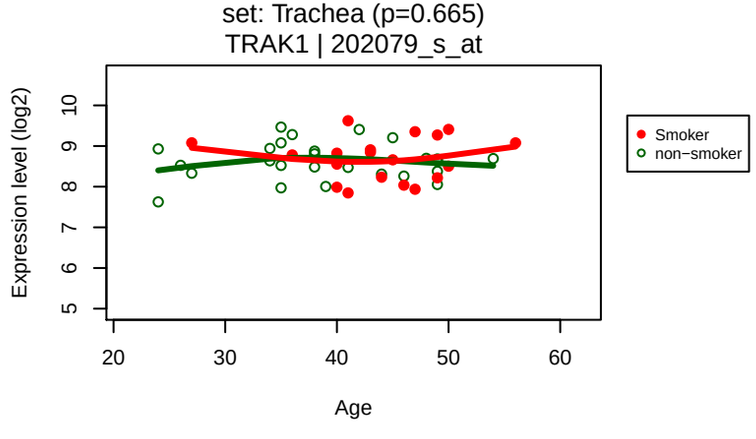

Gene expression ratio in Males
gene: TRAK1 | 202079_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0491
p=0.0893
p=0.61
p=0.1234
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRAK1 | 202079_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8003
p=0.0276
p=0.3627
p=0.5511
SmallAirway2
LargeAirway2
Trachea
Macrophage2

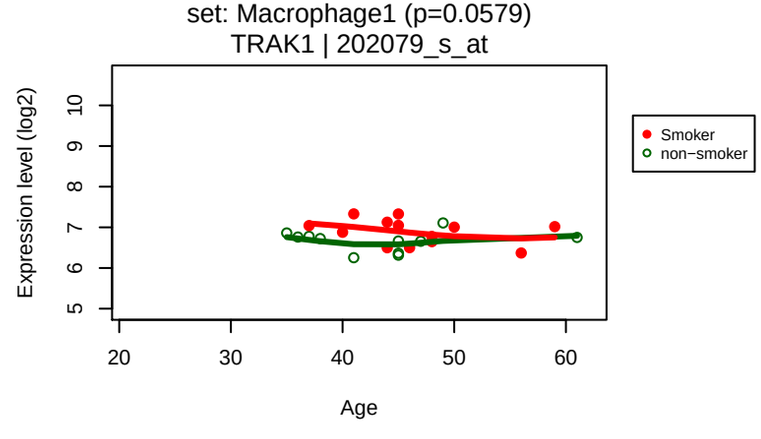

set: SmallAirway1 (p=0.0901)
TRAK1 | 202079_s_at
Expression level (log2)
Age
Smoker
non−smoker

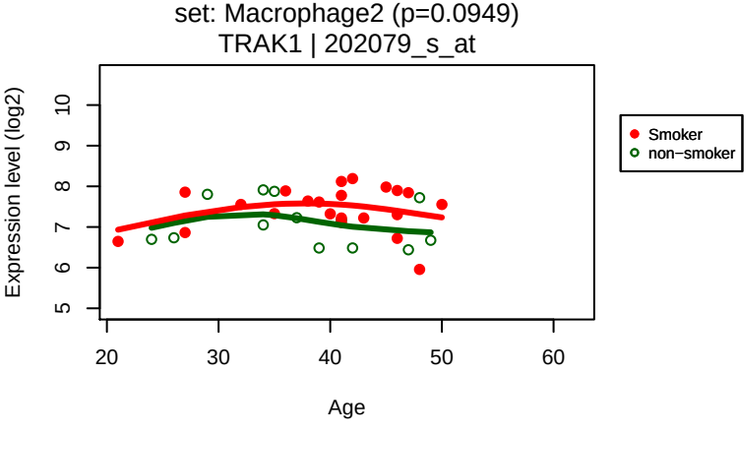

set: SmallAirway2 (p=0.0586)
TRAK1 | 202079_s_at
Expression level (log2)
Age
Smoker
non−smoker

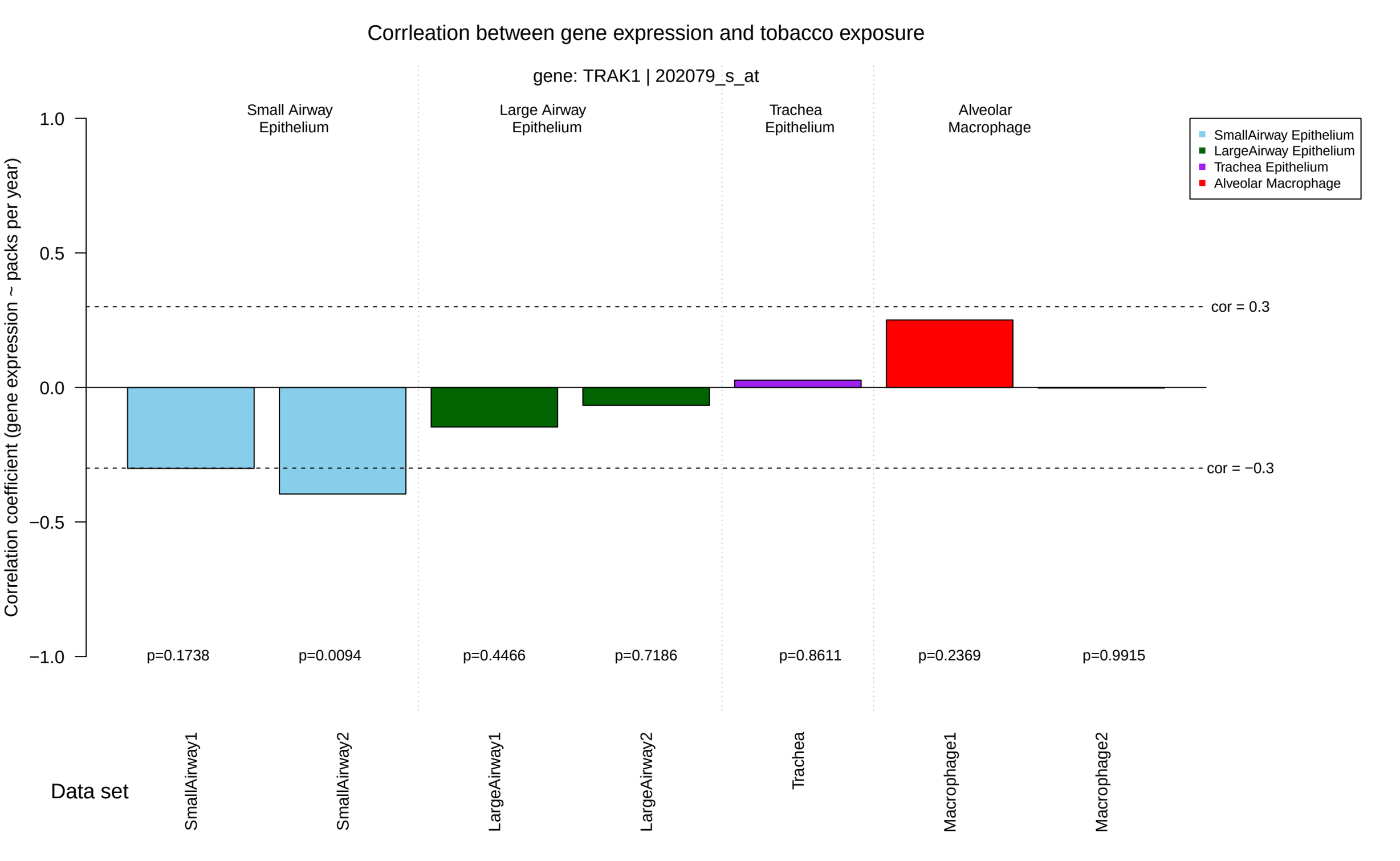

set: LargeAirway1 (p=0.0972)
TRAK1 | 202079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7325)
TRAK1 | 202079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.665)
TRAK1 | 202079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0579)
TRAK1 | 202079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0949)
TRAK1 | 202079_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRAK1 | 202079_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1738
p=0.0094
p=0.4466
p=0.7186
p=0.8611
p=0.2369
p=0.9915
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2